

TIPS TO AVOID COMMON DESIGN CHALLENGES

Hardware does not forgive assumptions. Challenging common misconceptions, exposing real design mistakes, and following disciplined design practices can prevent costly PCB re-spins.



CHETAN SONI
is Senior
Engineer at NXP
Semiconductors

We will explore the real challenges in hardware design and what it takes to get a board working correctly the first time. A common belief in the industry is that the first PCB will never work and that a second or even third revision is inevitable. Many engineers openly admit that; even before completing their first design, they expect a re-spin. This mindset has become commonplace, and it is exactly what we need to question.

The aim is to challenge that assumption and focus on a few fundamental practices that can significantly improve the chances of getting a design right the first time. To keep this engaging, we are joined by our very own Mr PCB, who will stay with us throughout and share practical design insights on the decisions that matter most.

Role of a hardware design engineer

Let us first understand who a hardware engineer or a hardware design engineer really is. Often, we imagine a hardware engineer as someone sitting at a desk in front of a PC, surrounded by messy wires, PCBs, and a soldering station. While that image is familiar, it is not the complete

picture. A hardware engineer is primarily responsible for the following:

Collecting and understanding requirements. This includes gathering requirements from multiple teams, such as the systems team, mechanical team, software team, testing team, and sourcing team. Hardware engineers work closely with many cross-functional teams and must clearly understand inputs from all of them. Under this responsibility, the hardware engineer also:

- Creates a system architecture to meet and fulfil system requirements
- Performs component selection, which is a critical task since every system depends on the right choice of components based on functionality, availability, and constraints

Performing analysis and simulations. Once requirements and components are defined, the hardware engineer performs analysis and simulations for the system or hardware being developed. Under this responsibility, the hardware engineer also:

- Creates schematics and PCB layout
- Creates required documentation throughout the hardware design lifecycle, starting from requirement collection to having a PCB in hand, and eventually taking